

F51 — Investigation #2 (K-type grading $\times$ restriction grading orthogonality), driven to closure: COUPLED, not orthogonal. The $SO(3,1)$ boosts are noncompact (p, not k), so $P_{\text{restriction}}$ mixes K-types while H_B is diagonal in them $\rightarrow [H_B, P_{\text{restriction}}] \neq 0$. Confirms muon = single-object T190 (F50); the coupling IS the dual- ρ cross-side structure.

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F51 — Investigation #2: Grading Coupling

0. Result

The K-type grading and the restriction grading on $H^2(D_{IV}^5)$ are COUPLED, not orthogonal: $[H_B, P_{\text{restriction}}] \neq 0$. The reason is clean and structural — the $SO(3,1)$ boosts are noncompact, so the restriction projection mixes the very K-types the Casimir grades. This confirms F50's muon result (no clean K-type $\otimes$ restriction factorization $\rightarrow$ the additive Composite v0.5 is illegitimate $\rightarrow$ muon = single-object T190), and it tells Investigation #3 that genuine cross-side observables exist.

1. The two operators

- H_B = Casimir of $K = SO(5) \times SO(2)$. Diagonal in K-types: $H_B|V_\lambda\rangle = C(\lambda)|V_\lambda\rangle$ (the K-Casimir, $\rho_{SO(5)}$ convention, F47).
- $P_{\text{restriction}}$ = projection per Casey #14, the substrate sequence $SO(5,2) \rightarrow SO(4,2)$ ($1/n_C$ chirality projection) $\rightarrow SO(3,1)$ (Casey #8 SCMP τ -direction). It projects H^2 onto the $SO(3,1)$ (Minkowski) content.

Orthogonal gradings $\nexists [H_B, P_{\text{restriction}}] = 0$.

2. The structural argument: boosts are noncompact $\rightarrow$ coupling

Decompose the $SO(3,1)$ Lorentz generators inside $SO(5,2)$: - 3 rotations J_i — compact, lie in $SO(3) \subset SO(5) \subset K$. These commute with the K -grading structure. - 3 boosts K_i — noncompact. In the Cartan decomposition $\mathfrak{so}(5,2) = \mathfrak{k} \oplus \mathfrak{p}$ ($\mathfrak{k} = \mathfrak{so}(5) \oplus \mathfrak{so}(2)$, $\mathfrak{p}$ the 10-dim noncompact part), the Lorentz boosts sit in $\mathfrak{p}$, not $\mathfrak{k}$.

The noncompact $\mathfrak{p}$ decomposes as $\mathfrak{p}^+ \oplus \mathfrak{p}^-$ on the Hermitian symmetric domain — these are exactly the K -type raising/lowering operators that build the holomorphic discrete series H^2 (Harish-Chandra). So any operator involving the boosts moves between K -types.

$P_{\text{restriction}}$, projecting onto $SO(3,1)$ -content, necessarily involves the boost directions ($SO(3,1)$ is generated by rotations *and* boosts). Therefore $P_{\text{restriction}}$ has a K -type-changing part (from the boosts in $\mathfrak{p}^+ \oplus \mathfrak{p}^-$). Since H_B is diagonal in K -types with distinct eigenvalues $C(\lambda)$,

$$[H_B, P_{\text{restriction}}] = \sum_{\lambda \neq \mu} (C(\lambda) - C(\mu)) (P_{\text{restriction}})_{\lambda\mu} |V_\lambda\rangle\langle V_\mu| \neq 0,$$

because the boost (off-diagonal, K -type-changing) matrix elements $(P_{\text{restriction}})_{\lambda\mu}$ are nonzero for $\lambda \neq \mu$ and the Casimir gaps $C(\lambda) - C(\mu) \neq 0$. The gradings are coupled.

3. Consequences

(a) Confirms F50 $\mu\text{on} \rightarrow \text{T190}$. Coupled gradings mean H^2 does not split as K -type $\otimes$ restriction. So the muon mass operator cannot be written as (K -type spectral term) + (orthogonal restriction term) — the additive Composite v0.5 form $(1575/8 + 81/8)$ is illegitimate under Reading (a). The muon is a single H^2 object $\rightarrow \text{T190} = (24/\pi^2)^{\{C_2\}}$. F50's structural line (ii) is now the explicit commutator argument. Walls 2/3 close: $\mu\text{on} = \text{T190}$; Composite v0.5 additive RETIRED.

(b) The coupling IS the dual- ρ cross-side structure (Investigation #3). F49's dual- ρ split (compact $\rho_{SO(5)} \leftrightarrow K$ -type/spectral; conformal $\rho \leftrightarrow \text{Plancherel/restriction-bulk}$) does not cleanly partition every observable, precisely because the two graded structures don't commute. The boosts that couple them are the same noncompact $\mathfrak{p}$ that the conformal ρ "sees." So genuine cross-side observables exist — observables whose value depends on the boost-coupling between the K -type and restriction structures. Investigation #3's classification must have a real "cross-side" column, not just two clean sides. (This also means the (1,1) bridge is not a free decoupling — the sides genuinely interact via $\mathfrak{p}$. Still a candidate-lead per K231c; the coupling is a reason the bridge is nontrivial, not yet a derived mechanism.)

(c) The muon and CKM divergence (F50) is grounded. The muon factorization fails (coupled gradings, this note); the CKM factorization succeeds as two-point in color (F50). Different spaces, opposite outcomes — and now both have explicit structural reasons.

4. Honest tiering + handoff

- Structural (solid): boosts $\in \mathfrak{p}$ (noncompact) $\nRightarrow$ $P_{\text{restriction}}$ mixes K-types $\nRightarrow [H_B, P_{\text{restriction}}] \neq 0 \nRightarrow$ COUPLED. This is standard Hermitian-symmetric rep theory ($\mathfrak{p}^+ \oplus \mathfrak{p}^-$ are K-type shifts); the verdict does not depend on the exact $P_{\text{restriction}}$ normalization.
- Needs Elie numerical: the explicit commutator *eigenvalue structure* on $V_{(1/2,1/2)}$, $V_{(3/2,1/2)}$, $V_{(5/2,1/2)}$ — magnitude of the coupling, which boost matrix elements dominate. Corroborates and quantifies; the coupled *verdict* is already structural.
- Confirms: F50 muon $\rightarrow$ T190 (Composite v0.5 additive retired). $\rightarrow$ @Keeper K232 (muon = T190) + K237 (gradings coupled, this note). $\rightarrow$ @Elie numerical $[H_B, P_{\text{restriction}}]$ on the three K-types. $\rightarrow$ Investigation #3 (Grace): add a genuine cross-side column.
- Tier: F51 v0.1 Investigation #2 closure (structural COUPLED); confirms muon T190; Elie numerical for eigenvalue magnitudes; cross-side structure flagged for #3.

5. Closure

Investigation #2 closes: the K-type and restriction gradings on H^2 are coupled, not orthogonal, because the $SO(3,1)$ boosts are noncompact (in $\mathfrak{p}$) and therefore mix the K-types that H_B grades — $[H_B, P_{\text{restriction}}] \neq 0$ by the $\mathfrak{p}^+ \oplus \mathfrak{p}^-$ K-type-shift structure. This confirms F50's muon result (no clean factorization $\rightarrow$ single-object T190, additive Composite v0.5 retired) and tells Investigation #3 that the dual- ρ classification has genuine cross-side observables (the boost-coupling), not two cleanly-separable sides. The exact eigenvalue magnitudes are Elie's numerical step; the coupled verdict is structural and solid. Investigations #1 and #2 now close together: CKM two-point (color), muon single-object T190 (coupled gradings) — opposite factorization outcomes, both grounded.

— Lyra, Investigation #2 closure. $[H_B, P_{\text{restriction}}] \neq 0 \rightarrow$ gradings COUPLED (not orthogonal). Structural: $SO(3,1)$ boosts are noncompact (Cartan $\mathfrak{so}(5,2)=\mathfrak{k} \oplus \mathfrak{p}$; boosts $\in \mathfrak{p}$, not $\mathfrak{k}=\mathfrak{so}(5) \oplus \mathfrak{so}(2)$); $\mathfrak{p}^+ \oplus \mathfrak{p}^-$ are the K-type raising/lowering ops building H^2 (Harish-Chandra holomorphic discrete series); $P_{\text{restriction}}$ onto $SO(3,1)$ -content involves boosts $\rightarrow$ mixes K-types; H_B diagonal in K-types with gaps $C(\lambda)-C(\mu) \neq 0 \rightarrow$ commutator $= \Sigma(C(\lambda)-C(\mu))(P_{\text{restriction}})\lambda\mu |V_\lambda\rangle\langle V_\mu| \neq 0$. *Consequences:* (a) confirms F50 muon $\rightarrow$ T190 (no K-type $\otimes$ restriction factorization $\rightarrow$ additive Composite v0.5 illegitimate $\rightarrow$ single-object T190; Walls 2/3 CLOSE, additive RETIRED); (b) coupling IS the dual- ρ cross-side structure (Investigation #3 needs a genuine cross-side column; boosts couple compact- ρ K-type side to conformal- ρ restriction side; (1,1) bridge nontrivial but still candidate-lead K231c); (c) muon/CKM divergence (F50) grounded — both factorization outcomes have explicit structural reasons. Elie numerical: exact commutator eigenvalues on $V_{(1/2,1/2)}$, $V_{(3/2,1/2)}$, $V_{(5/2,1/2)}$. Keeper K237 (coupled) + K232 (muon T190). Structural verdict solid (independent of $P_{\text{restriction}}$ normalization); numerics

quantify.